

SYSTEM-H blocker & NATQ-002 BM25 control

SYSTEM-H not executable · 0 validation runs consumed · BM25 control scored on 40 validation cases · 17/17 integrity gates · head `a9534f0` · 2026-09-04

Two results here, and neither is what it might look like.

SYSTEM-H has no result and did not fail. Its status is `CANNOT_EXECUTE_MISSING_SCORE_DETERMINING_ARTIFACTS` and `SUPPORTED` is **null, not false**. This is an execution and provenance blocker. Nothing about SYSTEM-H's quality was measured, and 0 of its preregistered validation runs were consumed.

The 0.375 below is BM25's number, not SYSTEM-H's. It is a control measurement with no pass/fail threshold. The approved floors belong to SYSTEM-H and were deliberately not applied to a comparator.

0	null	0.375	0	17/17
SYSTEM-H RUNS CONSUMED	SYSTEM-H SUPPORTED	BM25 CASE_HIT@10	RESERVE LOG BYTES	INTEGRITY GATES

1 — Why SYSTEM-H cannot be scored

SYSTEM-H-V2-DEV-CANDIDATE is an architecture *record*, not runnable code. Nine requirements were classified; eight are transfer problems, and one is not.

#	Requirement	Classification
1	Executable SYSTEM-H runner	MISSING_LOCAL · NEEDS_TRANSFER
2	Exact SYSTEM-H configuration record	AVAILABLE_LOCAL
3	SYSTEM-A-GLOBAL implementation and artifacts (stage 1, A_pool=100)	MISSING_LOCAL · NEEDS_TRANSFER
4	SYSTEM-G / EXP-019A score-determining implementation and artifacts	PROVENANCE_GAP · MISSING_LOCAL · NEEDS_TRANSFER
5	Projection index <code>ps_v2_ovl_win448_s224</code>	MISSING_LOCAL · NEEDS_TRANSFER
6	Cross-encoder model and tokenizer artifacts	MISSING_LOCAL · NEEDS_TRANSFER
7	All candidate-generation implementation hashes	MISSING_LOCAL · NEEDS_TRANSFER
8	Performance-path implementation (PERF-003 D1)	MISSING_LOCAL · NEEDS_TRANSFER
9	Everything needed to recompute <code>score_determining_hash</code>	PROVENANCE_GAP

Item 4 is the one copying cannot fix. The EXP-019A retrieval prior is *score-determining*, and its originals are absent from every fetched ref — `PROVENANCE-GAP-001` holds them only as relayed conversation values. Its behaviour *is* described well enough to re-implement, and that is precisely the trap: a re-implementation would run and produce numbers that could never be verified against the thing they claim to reproduce, because that thing does not exist.

The preregistration makes a run valid only if `score_determining_hash` equals `3fade96a9b3597861b0788c15d0e456a52fa92f6907749a2661960e54bf7b2e5` at run time, and that hash covers fields that would be invented rather than reproduced. Reproduction status: **False**. No field was invented to force a match, and no new configuration was given SYSTEM-H's identity.

No requirement needs a CUDA host. SYSTEM-H is CPU-executable in principle — it is provenance, not hardware, that blocks it. Two legitimate paths remain, and reconstruction is not one of them: a rebuild from the written description takes a new identity and inherits none of SYSTEM-H's claims.

2 — Threshold decision, recorded without touching the preregistration

Decision **APPROVED_PRE_RUN** · thresholds changed **False** · validation runs consumed before the decision **0**.

Floor	Value	Scope / definition / population
case_hit@10 PASS	0.8	unchanged
case_hit@10 FAIL	0.65	unchanged

These values are preregistered engineering acceptance floors, not empirically estimated performance priors. Their purpose is to prevent a system from qualifying on paired improvement while remaining below an engineeringly useful absolute performance level. Statistical/comparative evidence is provided separately by the preregistered paired test against the co-run BM25 baseline.

The decision is a **separate artifact referencing the preregistration by SHA** (`0e18154ea46d8d03da41e28ec54de88c5c4632c625084a71f8c1f867ed37b0f2`) rather than an edit to it. That keeps the preregistration's own hash and its “written before any scoring” claim independently checkable — editing it to improve the rationale would have destroyed exactly the property that makes it evidence. Preregistration unmodified: **True**.

3 — BM25 control (SYSTEM-BM25-NATQ-CONTROL)

Pure BM25 over `cs_v1_control`, $k1=1.2$, $b=0.75$, depth 10, randomness: none anywhere in this system or this evaluation; no seed is required and none is used. Scored on the **validation partition only** — 40 cases, 57 gold spans, 400 ranked rows persisted.

Excluded by construction: dense retrieval, projection retrieval, local W2O expansion, reranking, cross-encoder, SYSTEM-H blend, generated query expansion, facets, document conditioning, identifier matching, RRF fusion of any kind.

Metric	Value
case_hit@10	0.3750
micro_MRR	0.1425
case_full_coverage@10	0.3250
span_recall@10	0.3333
case_hit@1	0.1000
latency p50 / p95	394 ms / 634 ms

Per authoring slice

Slice	Cases	Spans	case_hit@10	micro_MRR	span_recall@10
A	8	13	0.375	0.0571	0.2308
B	8	12	0.500	0.3194	0.5000
C	8	11	0.250	0.0556	0.2727
D	8	10	0.375	0.2111	0.3000
E	8	11	0.375	0.0750	0.3636

A weak floor is the useful outcome here. 0.375 is the expected shape for pure BM25 against conversational questions, and it matters for two reasons: it leaves the future paired test real headroom to detect an effect, and it confirms the approved 0.80 floor is a demanding bar rather than one a keyword baseline already clears. Decision recorded: **None** — Control measurement. No pass/fail threshold applies to BM25.

The per-case hit vector is persisted in fixed case-id order as the **immutable comparator input** for a future SYSTEM-H paired bootstrap over cases.

4 — Integrity

17 of 17 post-run gates pass. The benchmark, split, lock and preregistration hashes were verified *before* the run and re-verified *after* it, unchanged in both directions.

- NATQ-002-ACCEPTED-100.json unchanged after the run
- validation.json unchanged after the run
- reserve.json unchanged after the run
- reserve.lock.json unchanged after the run
- NATQ-002-SPLIT-001.json unchanged after the run
- NATQ-002-PREREGISTRATION-001.json unchanged after the run
- reserve access log still empty
- SYSTEM-H record unmodified
- SYSTEM-H runs consumed = 0
- SYSTEM-H SUPPORTED is null (not false)
- SYSTEM-H score_determining_hash NOT reproduced
- BM25 scored validation only
- BM25 consumed 0 SYSTEM-H runs
- BM25 has no pass/fail decision
- BM25 case vector covers exactly the 40 validation cases
- ranked output contains only validation case ids
- artifact hashes recorded in the run record match on disk

Pinned hashes

Artifact	SHA256
benchmark file	1450b9e1ba892d84d2ecab279cfa229b128a02108a4704b56393382b6c5bf741
benchmark canonical	d1f2b51f61a05c7652cd420c32bedb6b72b2ad8cfe14a43ab0fe3d49774c9d60
validation	6b7f3c90e2bfa58f244de6b2aff65e56ca3f50e2ed0886e83696aba8f5b47961
reserve	14aaa80224b5ad4bea021ccc63ffc96d463e54c9ee71f27b552aaecc06c46e32
reserve lock	f60eca02d4223770d09146f250685f23a644fc35980e30178d54f0a1b0274215
reserve access log	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
systemh prereg	0e18154ea46d8d03da41e28ec54de88c5c4632c625084a71f8c1f867ed37b0f2
threshold decision	b0a9f504e682174e4de83d030d377d91c1b69b93612579fe7f9a0cf27a868efb
bm25 prereg	bb3c2b29866d40262c427aeaac4db0374e39caf2642570268160f2610dc4e5bc
bm25 source	b290848a8753d4a0e2dfafa4c752d2f0f6eae71d8288be5c959524493ca43d45
bm25 runner	9f5485c88887c27c137c0dcc647baac411219fa783e6e335aa0d6eef5be30e15
bm25 case results	061e475b5ed7df140919fd5ad60fb96d6ef31b0a64d39aac561fac31ee4a7054
bm25 ranked output	1eb9ffe15e3974f195da6a3145da8f8952a203658e9ed11ec1d2ad678a61ba6a
unblock requirements	582e11ec849217b142c8b002868088c7162ee10f9af950c925151936cf3a3cdc

5 — Constraints held

Constraint	State
system h reconstructed	False
system h modified	False
system h run	False
system h marked failed	False
system h score fabricated	False
reserve opened	False
split altered	False
bm25 tuned	False
bm25 given fake threshold	False
metric definitions invented	False

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